A recoverability boundary in a minimal rate-limited dynamical system

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Abstract

We study a minimal dynamical system subject to transient forcing and finite recovery capacity. We show that system behavior separates into return and non-return regimes defined by an operational boundary in the joint space of input load and forcing timescale. While the minimum state reached depends primarily on total input (AUC), recoverability depends on both load and timescale, and recovery time within the recoverable regime appears to be primarily controlled by forcing duration.

Additionally, we find that prior exposure can shift system behavior across this boundary, such that identical primary forcing may lead to different outcomes depending on system history.

These results suggest that failure may arise from rate mismatch rather than absolute load. In particular, we observe a separation of control parameters across observables: excursion depth depends primarily on total input, recoverability depends jointly on load and timescale, and recovery dynamics are primarily governed by forcing duration. Additionally, recoverability may depend not only on instantaneous forcing conditions, but also on the system's prior trajectory.

1 Introduction

Failure in complex systems is often associated with exceeding absolute limits such as energy thresholds or load capacity. However, systems may fail even when total input remains within nominal bounds, suggesting that temporal structure plays a critical role.

Recent work in dynamical systems has emphasized the role of rate-induced tipping (R-tipping), in which system transitions depend on the rate of forcing rather than its absolute magnitude. Such behavior has been studied in ecological, climate, and physical systems, and has been linked to early-warning signals and critical rate thresholds, as well as recent developments in minimal solvable models.

In recovery-limited systems, behavior may depend on the relationship between input timescales and intrinsic recovery dynamics. Under transient forcing, identical total input may lead to different outcomes depending on how rapidly it is delivered.

This perspective is consistent with rate-induced tipping frameworks, in which system outcomes may depend on the rate of forcing relative to intrinsic system timescales.

Here, we investigate this question using a minimal dynamical model. By avoiding system-specific assumptions, we isolate structural features of recovery-limited dynamics.

We show that system behavior separates into distinct regimes defined by an operational boundary in the joint space of input load and forcing timescale, and that different observables respond to different control parameters.

2 Model

We consider a minimal recovery-limited dynamical system subject to transient forcing. The system is described by two coupled state variables:

$$\frac{dv}{dt} = \frac{1-v}{\tau_r} - d \tag{1}$$

$$\frac{dd}{dt} = \alpha\, u(t) - \beta\, d \tag{2}$$

where $v(t)$ represents a proxy for system recoverability or integrity, and $d(t)$ represents accumulated effective damage.

The parameter τ_r is the intrinsic recovery timescale of the system. The parameter α controls the rate at which forcing contributes to damage accumulation, while β sets the rate of damage dissipation or repair.

The recovery of $v(t)$ depends indirectly on the dissipation of accumulated damage $d(t)$, linking system recovery to finite-rate internal processes.

The forcing $u(t)$ is modeled as a Gaussian pulse with fixed area (AUC), defined as:

$$u(t) = \frac{\mathrm{AUC}}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(t-t_0)^2}{2\sigma^2}\right) \tag{3}$$

where σ controls the pulse width (forcing timescale), and t_0 is the pulse center. By construction, the integral of $u(t)$ over time equals AUC, allowing independent control of total input and delivery timescale.

Simulations are performed over a finite time window, and system behavior is characterized using the following operational definitions:

- **Collapse**: the system is considered to collapse if $v(t)$ falls below a predefined threshold v_{collapse} at any time.
- **Return**: the system is considered to return if $v(t)$ recovers above a threshold v_{return} within a fixed observation window.
- **Recovery time**: defined as the first time at which $v(t)$ exceeds v_{return}, conditioned on return.

In practice, the primary regime classification is based on return versus non-return within the observation window, while collapse and recovery time are used as complementary operational descriptors.

We explore system behavior across a parameter grid spanning AUC and pulse width, allowing us to map regions of return and non-return and to quantify recovery dynamics within the recoverable regime.

Code and reproducible simulations are available at:
`https://github.com/jaimeojse-collab/oxygen-regime-boundary-minimal`

3 Results

3.1 Recoverability boundary

We first examine system behavior across the parameter space defined by total input load (AUC) and forcing timescale (pulse width). For each parameter pair, we determine whether the system returns above a fixed threshold within a finite observation window.

The results suggest a well-defined operational boundary separating return and non-return regimes. This boundary is not determined by AUC alone: for a given total input, system outcome depends on how rapidly the input is delivered. Specifically, higher AUC values require slower forcing (larger pulse width) for recovery to remain accessible.

At low AUC, the system appears robust to variations in forcing timescale, with return observed across a broad range of widths. At higher AUC, however, recoverability becomes increasingly sensitive to input rate, and return is only observed when forcing is sufficiently slow.

These observations indicate that recoverability emerges from the joint effect of load and timescale, rather than from total input alone.

3.2 Separation of observables

To further characterize system behavior, we analyze three observables: the minimum state reached during the trajectory ($v_{\min}$), the binary return outcome, and the recovery time for trajectories that return.

We find that these observables respond differently to the control parameters:

- The minimum state reached ($v_{\min}$) depends primarily on AUC and shows only weak sensitivity to pulse width. Across the parameter space, $v_{\min}$ varies smoothly with total input, indicating that excursion depth is largely load-driven.
- In contrast, the return outcome exhibits a sharp transition in the joint space of AUC and pulse width, forming the recoverability boundary described above. This indicates that recoverability is controlled by a combination of load and timescale.
- Within the subset of trajectories that return, recovery time appears relatively insensitive to AUC but varies systematically with pulse width. For a given width, recovery times remain approximately constant across AUC values within the recoverable regime, while increasing width leads to longer recovery times.

Taken together, these results suggest a separation of control parameters across observables: load primarily determines excursion depth, while timescale plays a dominant role in recovery dynamics, and both jointly determine recoverability.

This separation suggests that distinct observables encode different aspects of system dynamics, rather than representing redundant measures of the same underlying process.

3.3 Recovery time within the recoverable regime

We next examine recovery time as a function of AUC for different pulse widths, restricting attention to trajectories that return.

Across a range of pulse widths, recovery time curves are approximately flat with respect to AUC, indicating weak dependence on total input within the recoverable regime. However, these curves are clearly separated across pulse widths, with larger widths corresponding to longer recovery times.

This pattern suggests that, once recovery remains accessible, the duration of recovery is governed primarily by the timescale of the input rather than by its total magnitude. In other words, the system retains memory of the temporal structure of the forcing even when the total input is similar.

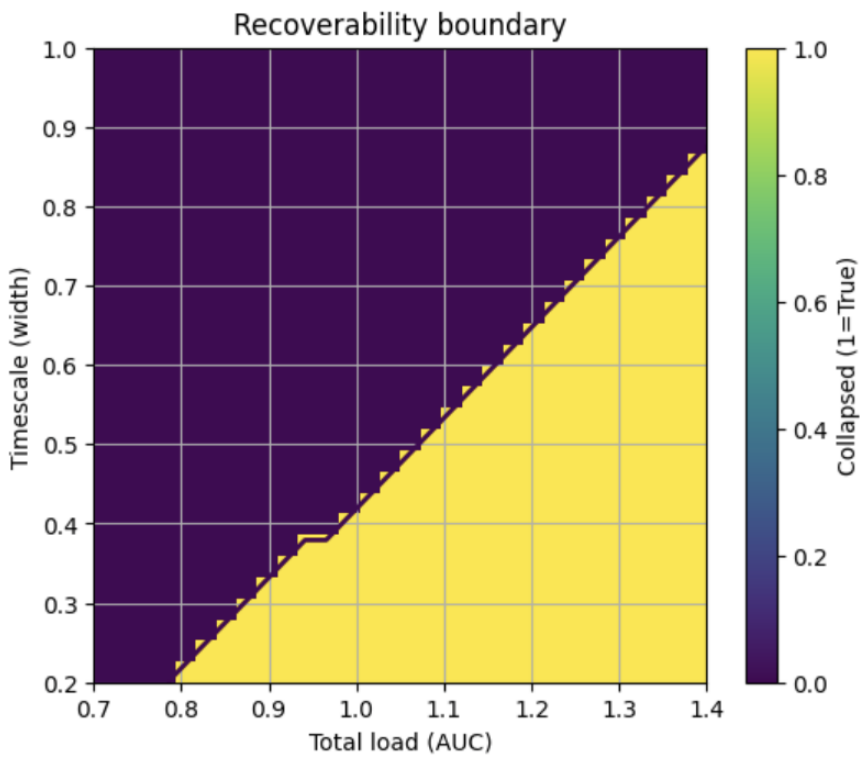

Figure generated from simulations in the linked repository.

Figure 1: Recoverability boundary in (AUC, timescale) space. The system exhibits a transition between recoverable and non-recoverable regimes that depends jointly on total load and forcing timescale.
Figure generated from simulations in the linked repository.

3.4 Robustness across parameters

To assess robustness, we repeat the analysis across a range of model parameters controlling damage accumulation and recovery (α, β). While the precise location of the recoverability boundary shifts with parameter values, its qualitative structure remains intact.

In all tested cases, we observe:

- a transition between return and non-return regimes in the joint parameter space,
- a smooth dependence of $v_{\min}$ on AUC,
- and a separation of recovery time curves by pulse width.

These results suggest that the emergence of a recoverability boundary and the separation of observables are not artifacts of specific parameter choices, but rather reflect a structural property of the model.

3.5 History-dependent shifts in recoverability

To examine whether recoverability depends solely on current forcing conditions or also on prior system state, we perform a double-pulse experiment. The system is first exposed to a pre-conditioning

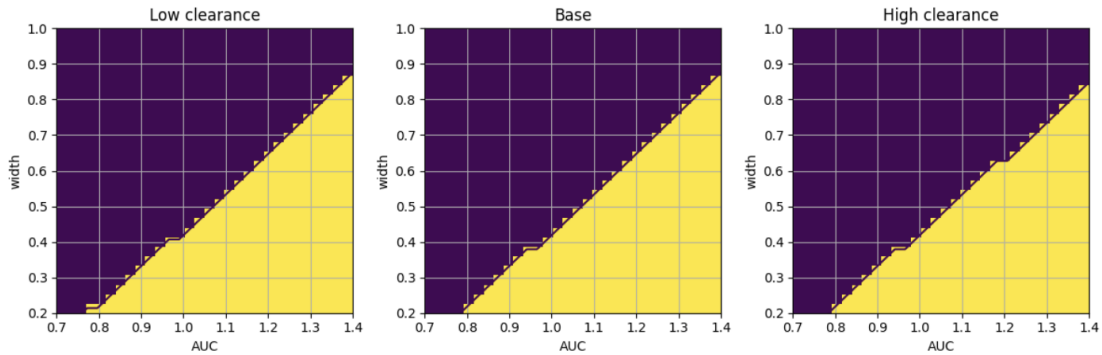

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Figure 2: Robustness of the recoverability boundary under parameter variation. While the boundary shifts with parameter values, its qualitative structure remains consistent.
Figure generated from simulations in the linked repository.

pulse, followed by a primary pulse identical to that used in the single-pulse experiments.

We compare trajectories with and without prior exposure while keeping the primary forcing identical. In the absence of a pre-pulse, the system remains within the recoverable regime and returns above threshold within the observation window.

However, when preceded by a subcritical pre-pulse, the same primary forcing leads to collapse, with the system crossing below the collapse threshold before recovery becomes accessible.

This result suggests that prior exposure can move the system across the operational recoverability boundary identified in the single-pulse analysis, even when the primary forcing remains unchanged. The transition between recoverable and non-recoverable regimes is therefore not determined solely by instantaneous forcing parameters, but may depend on the system's trajectory through state space.

Mechanistically, the pre-pulse transiently increases accumulated damage and reduces usable recovery capacity, altering the effective initial condition for the subsequent perturbation.

These observations indicate that the effective position of the recoverability boundary depends on the system state, and that system outcomes may depend on whether a given forcing regime is entered from a recoverable or compromised state., and that system outcomes may depend on whether the system enters a given forcing regime from a recoverable state.

4 Interpretation

The results presented here suggest that system behavior in recovery-limited dynamics may not be governed solely by total input, but by the interaction between input magnitude and input timescale. In particular, the emergence of a recoverability boundary indicates that accessibility of recovery depends on whether the system can respond within a finite time window to a given forcing profile.

One interpretation of this behavior is that the system is constrained not only by the amount of perturbation it experiences, but by the rate at which it can process or dissipate that perturbation. In this sense, failure may arise from a mismatch between the timescale of the input and the intrinsic recovery timescale of the system.

The separation observed between different observables further supports this view. While the depth of the excursion ($v_{\min}$) appears primarily determined by total input, the ability to return depends on both load and timescale, and the recovery dynamics within the recoverable regime

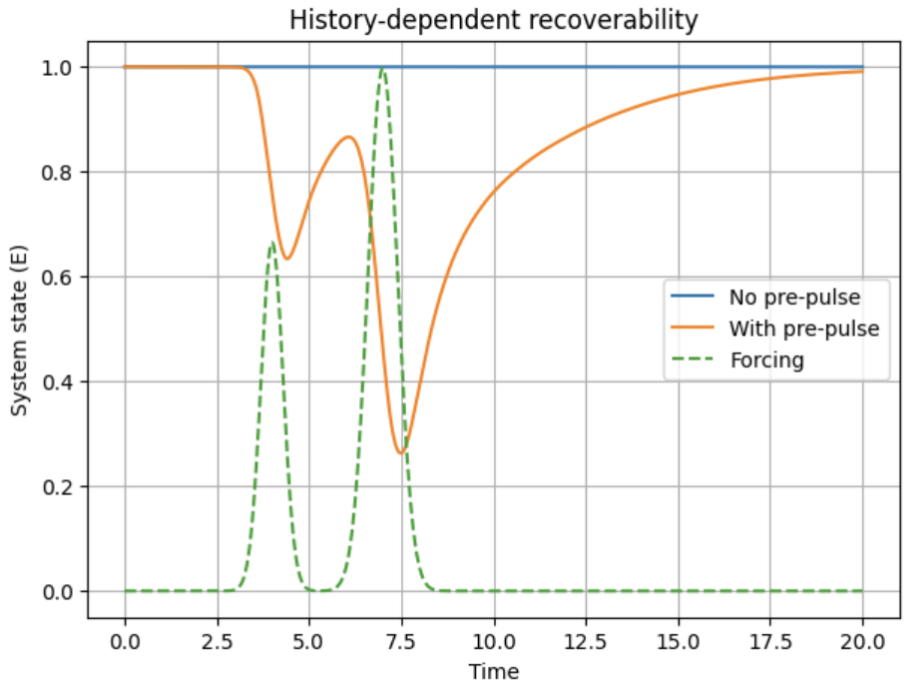

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Figure 3: History-dependent recoverability in a double-pulse experiment. Under identical primary forcing, the system remains recoverable without prior exposure but collapses when preceded by a subcritical pre-pulse, indicating dependence on system history.
Figure generated from simulations in the linked repository.

appear dominated by forcing duration. This decoupling suggests that different aspects of system behavior are controlled by distinct mechanisms or constraints.

Although the model is intentionally minimal, such behavior may be consistent with systems in which recovery capacity is not instantaneous, but mediated by internal processes with finite rates. In biological contexts, for example, recovery may depend on the ability to convert or redistribute resources across states. Under such conditions, the total resource pool may be less relevant than the rate at which usable states can be accessed.

More generally, these results suggest that failure in recovery-limited systems may be better understood as a loss of access to functional states over time, rather than as a simple exhaustion of capacity.

This separation of control parameters across observables extends the perspective of rate-induced tipping frameworks, where emphasis is typically placed on transition conditions. Here, we show that distinct observables may be governed by different control parameters within a recovery-limited regime.

This behavior is analogous to recent solvable minimal models for rate-induced tipping, such as Yoshioka (2026), though here we emphasize the separation of control parameters across multiple observables in a recovery-limited context.

The observation of history-dependent transitions further suggests that recoverability boundaries may not be fixed in parameter space, but may shift depending on prior system state. In this sense, failure may arise not only from rate mismatch, but from entering a given perturbation regime from a compromised state in which recovery capacity has already been partially degraded.

In biological systems, similar constraints may be consistent with processes such as NAD^+ regeneration, where dysfunction may not reflect depletion of total substrate, but rather limitations in the system's ability to regenerate functional states under sustained demand.

In this sense, system failure may emerge when regeneration dynamics fail to keep pace with ongoing metabolic load, consistent with a recovery-limited interpretation of system behavior.

This perspective suggests that effective system limits may be governed not by total resource availability, but by the rate at which functional states can be restored over time.

5 Limitations

The model considered here is deliberately minimal and does not aim to capture the full complexity of real-world systems. In particular, it relies on a single state variable and a simplified representation of forcing and recovery.

Several limitations should be noted. First, the model does not include explicit mechanisms underlying recovery processes, and therefore does not identify specific physical, biological, or chemical pathways. Second, the analysis is based on numerical simulations and does not include a formal analytical treatment of the observed boundary. Third, the results have not been directly compared to experimental or observational data.

As a consequence, the findings should be interpreted as identifying a possible structural feature of recovery-limited systems, rather than as providing a complete description of any specific system.

Future work may extend this framework by incorporating additional internal variables, exploring analytical approximations, and comparing model predictions with empirical data from systems exhibiting recovery-limited dynamics.

6 Conclusion

We identify a recoverability boundary in a minimal system under transient forcing and show that system behavior separates into return and non-return regimes in the joint space of input load and forcing timescale.

Beyond this boundary, we observe a separation of control parameters across observables: excursion depth is primarily determined by total input, recoverability depends jointly on load and timescale, and recovery dynamics within the recoverable regime are governed predominantly by forcing duration. This indicates that distinct observables encode different aspects of system dynamics rather than reflecting a single underlying constraint.

These results suggest that failure in recovery-limited systems may arise from a mismatch between the timescale of the input and the intrinsic recovery timescale of the system, rather than from absolute load alone.

We further show that prior exposure can alter system outcomes under identical forcing conditions, indicating that the effective accessibility of recovery depends on system state. This suggests that recoverability boundaries may be state-dependent, and that system history may influence whether recovery remains dynamically accessible.

Taken together, these findings support a view in which system limits are governed not only by total resource availability, but by the rate at which functional states can be restored over time.

References

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